

Project I- House Price Prediction

In this project, I worked on a housing dataset to predict house prices using machine learning. I started by exploring and cleaning the data, checking for missing values and duplicates, and converting categorical columns into numerical values. I then performed exploratory data analysis using visualizations such as a house price distribution histogram, correlation heatmap, and scatter plot.

After splitting the dataset into training and testing sets, I trained two models: Linear Regression and Random Forest Regressor. The Linear Regression model performed better, achieving an R^2 score of 0.653 compared to 0.612 for Random Forest.

The analysis showed that factors such as area, number of bathrooms, stories, and air conditioning have a strong influence on house prices. One interesting observation was that the simpler Linear Regression model outperformed the more complex Random Forest model.

Overall, this project helped me understand data preprocessing, visualization, model building, and performance evaluation using real-world data.

Key Findings

- Linear Regression performed better than Random Forest, achieving an R^2 score of 0.653.
- Area, number of bathrooms, and number of stories were among the most important factors influencing house prices.
- Houses with better amenities, such as air conditioning and furnishing, generally had higher prices.
- The model was able to explain about 65% of the variation in house prices, providing reasonably accurate predictions for the dataset.